

ETH Zürich

Semester Project

CTabPFN: Conformal Uncertainty Quantification with TabPFN

Samuel Anzalone

Supervisors: Dr. Jakob Heiss

Prof. Dr. Josef Teichmann

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Samuel Anzalone
ETH Zurich

Jakob Heiss
University of California, Berkeley

Abstract

TabPFN approximates the Bayesian posterior predictive distribution of tabular data in a single forward pass, combining aleatoric and epistemic uncertainty by construction, yet its prediction intervals carry no coverage guarantee. We propose CTabPFN, three conformal calibrations of the native distribution, additive, multiplicative, and quantile-level, which all share a single forward pass and satisfy the marginal coverage guarantee. On 12 real datasets, CTabPFN restores the target coverage at nearly native interval width, roughly a quarter narrower than conformal methods that train their own models.

1 INTRODUCTION

Prediction intervals are one of the most natural ways to quantify the uncertainty of a regression model: rather than a point estimate, they provide bounds that contain the true response with a desired probability. Conformal prediction (Vovk et al., 2005) has emerged as the standard way to make such intervals trustworthy. Given n exchangeable points $(X_i, Y_i) \in \mathbb{R}^p \times \mathbb{R}$ and a miscoverage level α , it constructs an interval $\hat{C}_n(X_{n+1})$ satisfying the marginal coverage guarantee

$$\mathbb{P}\left(Y_{n+1} \in \hat{C}_n(X_{n+1})\right) \geq 1 - \alpha \quad (1)$$

with no assumptions on the data distribution or the underlying model.

The guarantee in (1) holds regardless of the model that produced the interval, but the quality of the interval does not. Sharp, locally adaptive intervals require a base predictor that captures the shape of $Y \mid X$, and conformal regression methods train one from scratch on every dataset: CQR (Romano et al., 2019) fits quantile regressors, while UACQR (Rossellini et al., 2024), PCS-UQ (Agarwal et al., 2025), and CLEAR (Azizi et al., 2026) fit large ensembles of bootstrapped models.

TabPFN (Hollmann et al., 2023, 2025; Grinsztajn et al., 2026) removes this training step. It is a transformer pretrained on synthetic datasets drawn from a prior over tabular data-generating processes. A single forward pass over the training points returns a full posterior predictive distribution of $Y_{n+1} \mid X_{n+1}$ for each test point X_{n+1} and any quantile of this distribution can be read off directly. Hence, TabPFN effectively is a quantile regressor that requires no fitting. Its intervals inherit the adaptivity of the posterior, widening and narrowing with the local uncertainty of the data. However, they carry no coverage guarantee. In our benchmark, at nominal 90% coverage, native TabPFN v3 (Grinsztajn et al., 2026) intervals can cover anything between 87.5% and 100.0% depending on the dataset.

We propose to conformalize the distribution that TabPFN already provides instead of training a new uncertainty model beside it. Since one forward pass exposes every quantile level, the main calibration schemes of the conformal literature can all be applied to the same shared inference. We call the resulting methods CTabPFN-A, CTabPFN-M, and CTabPFN-Q, standing for additive (Romano et al., 2019), multiplicative (Kivaranovic et al., 2020; Agarwal et al., 2025), and quantile-level (Chernozhukov et al., 2021) calibration of the native intervals. All three restore the guarantee (1) while preserving the adaptivity of the underlying posterior. While additive calibration of TabPFN has been tried (van Leeuwen, 2025), to our knowledge the multiplicative and quantile-level variants have not been published before.

Comparing these methods fairly is a measurement problem of its own. Published implementations differ in splits, seeds, library versions, and calibration conventions, and such deviations are easily mistaken for method effects. We therefore built `uq-bench`¹, a practical benchmark harness designed around reproducibility, encapsulation, and fairness, with 20 real-world regression datasets available to benchmark on. To the best of our knowledge, no standard benchmark for uncertainty quantification exists, and we release

¹<https://github.com/anzamel/uq-bench>

uq-bench as a step toward such a standard.

2 EXISTING METHODS

2.1 Split conformal prediction

Split conformal prediction (Vovk et al., 2005; Lei et al., 2018) divides the n points (X_i, Y_i) into a training set of size n_0 and a calibration set of size n_1 . In its basic form, a predictive model $\hat{\mu}$ is fit on the training set, and the interval $\hat{C}_n(X_{n+1}) = \hat{\mu}(X_{n+1}) \pm \hat{t}$ is returned, where $\hat{t}$ is the $\lceil (1 - \alpha)(n_1 + 1) \rceil$ -th smallest absolute residual $|Y_i - \hat{\mu}(X_i)|$ on the calibration set. This interval satisfies marginal coverage (1). However, its width is the same for every test point, namely $2\hat{t}$.

The construction generalizes to intervals of any shape through nested sets (Gupta et al., 2022). First, the method fits its predictors on the n_0 training points, for instance a mean regressor $\hat{\mu}$ or a pair of quantile regressors $\hat{q}_{\alpha/2}, \hat{q}_{1-\alpha/2}$. These fitted functions define an interval shape at each point x , e.g. $[\hat{q}_{\alpha/2}(x), \hat{q}_{1-\alpha/2}(x)]$ or the single point $\hat{\mu}(x)$. A family of intervals $\{\hat{C}_{n_0,t}(x)\}_{t \in \mathcal{T}}$, $\mathcal{T} \subseteq \mathbb{R}$, is then obtained by letting the index t widen or tighten this shape, e.g. $[\hat{q}_{\alpha/2}(x) - t, \hat{q}_{1-\alpha/2}(x) + t]$ or $\hat{\mu}(x) \pm t$. The intervals should be monotone in t , meaning $\hat{C}_{n_0,t}(x) \subseteq \hat{C}_{n_0,t'}(x)$ for $t < t'$. The calibration set then selects one member of the family,

$$\hat{t} = \inf \left\{ t : \sum_{i=n_0+1}^n \mathbf{1}_{Y_i \in \hat{C}_{n_0,t}(X_i)} \geq (1 - \alpha)(n_1 + 1) \right\}, \quad (2)$$

and the returned interval is $\hat{C}_n(X_{n+1}) = \hat{C}_{n_0,\hat{t}}(X_{n+1})$.

² Any interval of this form satisfies (1) (Vovk et al., 2005; Gupta et al., 2022). The beauty of this framework is that every method in this paper, existing and proposed, can be represented within it once its hyperparameters are fixed, differing only in the family $\{\hat{C}_{n_0,t}\}_{t \in \mathcal{T}}$.

2.2 A timeline of conformal regression

CQR Conformalized quantile regression (Romano et al., 2019) fits a pair of quantile regressors and expands them by a constant, $\hat{C}_{n_0,t}(x) = [\hat{q}_{\alpha/2}(x) - t, \hat{q}_{1-\alpha/2}(x) + t]$. The shape adapts to heteroscedasticity, but the calibration inflates every point by the same margin.

CQR-r and CQR-m Sesia and Candès (2020) and Kivaranovic et al. (2020) make the inflation it-

self adaptive by scaling it with the fitted interval. CQR-r scales t by the fitted interval width $\hat{w}(x) = \hat{q}_{1-\alpha/2}(x) - \hat{q}_{\alpha/2}(x)$, giving $\hat{C}_{n_0,t}(x) = [\hat{q}_{\alpha/2}(x) - t\hat{w}(x), \hat{q}_{1-\alpha/2}(x) + t\hat{w}(x)]$, while CQR-m replaces $\hat{w}$ on each side by the distance to the fitted median, $\hat{w}_{\text{lo}} = \hat{q}_{0.5} - \hat{q}_{\alpha/2}$ and $\hat{w}_{\text{hi}} = \hat{q}_{1-\alpha/2} - \hat{q}_{0.5}$.

DCP Distributional conformal prediction (Chernozhukov et al., 2021) fits the whole conditional quantile function and indexes the family by the quantile level itself, $\hat{C}_{n_0,t}(x) = [\hat{q}_{\tau_t}(x), \hat{q}_{1-\tau_t}(x)]$. Here t enters implicitly through the level τ_t . As t grows, the level decreases and the interval becomes larger.

UACQR Rossellini et al. (2024) argue that the inflation should scale with epistemic rather than aleatoric uncertainty, since the fitted quantile pair already accounts for the latter. To estimate it, they train an ensemble of B quantile regressors on bootstrapped subsets of the training data. UACQR-S scales t by the pointwise standard deviation $\hat{g}(x)$ of the ensemble’s quantile estimates, and UACQR-P indexes the family by their order statistics.

PCS-UQ Agarwal et al. (2025) build the shape from an ensemble. From a pool of candidate models, the top k by validation loss are selected, the selected models are refit on B bootstrapped datasets, and the pointwise median $\hat{m}$ and quantiles $\hat{l}, \hat{u}$ of the ensemble predictions define $\hat{C}_{n_0,t}(x) = [\hat{m} - t(\hat{m} - \hat{l}), \hat{m} + t(\hat{u} - \hat{m})]$. The multiplicative t widens intervals most where the ensemble disagrees.

CLEAR Azizi et al. (2026) decompose the uncertainty explicitly. An aleatoric estimate $\hat{q}^{\text{ale}}$ from quantile regression on residuals and an epistemic estimate $\hat{q}^{\text{epi}}$ from a PCS ensemble are combined as $\hat{C}_{n_0,t}(x) = [\hat{f} - t(\hat{q}_{\text{lo}}^{\text{ale}} + \lambda \hat{q}_{\text{lo}}^{\text{epi}}), \hat{f} + t(\hat{q}_{\text{hi}}^{\text{ale}} + \lambda \hat{q}_{\text{hi}}^{\text{epi}})]$, where the mixing ratio λ is a hyperparameter tuned by quantile loss. Their conformalized variant fixes λ on a separate split and retains the guarantee (1).

Current research concentrates on separating aleatoric from epistemic uncertainty and on recombining the two so that coverage holds not only marginally but also conditionally.

2.3 TabPFN for conformal regression

The Bayesian posterior predictive distribution combines aleatoric noise and epistemic model uncertainty by construction, in proportions that vary with x (Azizi et al., 2026). Let θ denote a data-generating process, i.e. a conditional distribution $p(y|x, \theta)$ of the response given the features. For a prior π over such processes and training data $D = \{(X_i, Y_i)\}_{i=1}^{n_0}$, the posterior pre-

²In practice, $\hat{t}$ is not found by searching over t . It is the $\lceil (1 - \alpha)(n_1 + 1) \rceil$ -th smallest of the per-point values $t_i = \inf\{t : Y_i \in \hat{C}_{n_0,t}(X_i)\}$, the conformity scores, which admit closed forms for all methods in this paper.

dictive distribution is

$$p(y|x, D) = \int p(y|x, \theta) p(\theta|D) d\theta, \quad (3)$$

where the likelihood $p(y|x, \theta)$ carries the aleatoric noise of the process θ and the posterior $p(\theta|D)$ carries the epistemic uncertainty about which process generated the data. The law of total variance makes the decomposition explicit,

$$\begin{aligned} \text{Var}(Y | x, D) &= \underbrace{\mathbb{E}_{\theta|D} [\text{Var}(Y | x, \theta)]}_{\text{aleatoric}} \\ &\quad + \underbrace{\text{Var}_{\theta|D} (\mathbb{E}[Y | x, \theta])}_{\text{epistemic}}. \end{aligned}$$

Where D constrains θ well, the epistemic term vanishes and the posterior predictive tightens to the noise level. Where it does not, plausible processes disagree and the posterior predictive widens. This combination requires no model-specific derivation and no ensemble assembly.

TabPFN approximates exactly the object in (3). Müller et al. (2022) provide the theoretical foundation and show that the pretraining loss of the network is a KL divergence to (3), minimized exactly by the posterior predictive $p(y|x, D)$ under the pretraining prior π . The forward pass over D thus is the Bayesian inference, with π fixed at pretraining time.

We note one limitation. The approximation holds under TabPFN’s own prior π , so a mismatch with the true process distorts the calibration level, even though the prior and the pretraining scale have improved with each version (Hollmann et al., 2023, 2025; Grinsztajn et al., 2026). Conformal calibration closes exactly this gap.

Despite this fit, the native distribution of TabPFN has hardly been conformalized. van Leeuwen (2025) applies the additive CQR score to the native quantiles in a missing-data setting, the only recorded instance we found. Other work wraps residual-based split conformal around TabPFN point predictions (Felizzi, 2026) or benchmarks the native distribution against separately built conformal baselines without conformalizing it (Landsgesell et al., 2026). To our knowledge, no published work applies multiplicative or quantile-level calibration to the native distribution of TabPFN or any tabular foundation model.

3 CONFORMALIZED TABPFN

3.1 Setup

As in Section 2.1, we split the n samples into n_0 training and n_1 calibration points. A single forward pass of TabPFN over the n_0 training points

returns, for every calibration and test point x , the native quantiles $\hat{q}_\tau(x)$ at every level τ of the grid $\mathcal{G} = \{0.001, 0.002, \dots, 0.999\}$, strictly increasing in τ . We write $\hat{m}(x) = \hat{q}_{0.5}(x)$ for the native median. All three methods below consume this one shared inference and TabPFN is never refit. The training fraction n_0/n is the single hyperparameter of our methods. We default to 0.9 and examine this choice in Section 5.

3.2 CTabPFN-A

CTabPFN-A applies the CQR score of Section 2.2 to the native quantiles,

$$\hat{C}_{n_0, t}(x) = [\hat{q}_{\alpha/2}(x) - t, \hat{q}_{1-\alpha/2}(x) + t], \quad t \in \mathbb{R}. \quad (4)$$

The conformity score of a calibration point is

$$t_i = \max \{ \hat{q}_{\alpha/2}(X_i) - Y_i, Y_i - \hat{q}_{1-\alpha/2}(X_i) \}, \quad (5)$$

and $\hat{t} = t_{(\lceil (1-\alpha)(n_1+1) \rceil)}$, where $t_{(k)}$ denotes the k -th smallest score on the calibration set. The correction is uniform in x . Every interval is expanded, or contracted for $\hat{t} < 0$, by the same margin, which is set by the largest calibration errors and can therefore be driven by a few extreme points.

Algorithm 1 CTabPFN-A

Require: $\{(X_i, Y_i)\}_{i=1}^n$, miscoverage level α

- 1: split n into n_0 training and n_1 calibration points
- 2: $\hat{q}_\tau(\cdot)$, $\tau \in \mathcal{G} \leftarrow$ TabPFN fitted through forward pass on n_0 training points
- 3: $t_i \leftarrow (5)$ for $i = n_0 + 1, \dots, n$ calibration points
- 4: $\hat{t} \leftarrow t_{(\lceil (1-\alpha)(n_1+1) \rceil)}$

Ensure: $\hat{C}_n(x) = [\hat{q}_{\alpha/2}(x) - \hat{t}, \hat{q}_{1-\alpha/2}(x) + \hat{t}]$

3.3 CTabPFN-M

CTabPFN-M calibrates multiplicatively around the native median,

$$\begin{aligned} \hat{C}_{n_0, t}(x) &= [\hat{m}(x) - t (\hat{m}(x) - \hat{q}_{\alpha/2}(x)), \\ &\quad \hat{m}(x) + t (\hat{q}_{1-\alpha/2}(x) - \hat{m}(x))], \quad t \geq 0. \end{aligned} \quad (6)$$

The conformity score of a calibration point is

$$t_i = \max \left\{ \frac{\hat{m}(X_i) - Y_i}{\hat{m}(X_i) - \hat{q}_{\alpha/2}(X_i)}, \frac{Y_i - \hat{m}(X_i)}{\hat{q}_{1-\alpha/2}(X_i) - \hat{m}(X_i)} \right\}, \quad (7)$$

and $\hat{t} = t_{(\lceil (1-\alpha)(n_1+1) \rceil)}$ as before. The score is a ratio, so the correction acts in relative units of the native interval. Each interval is scaled in proportion to its own width, and $\hat{t} < 1$ shrinks it below the native interval. The denominators are positive since the native quantiles are strictly increasing in τ .

Algorithm 2 CTabPFN-M

Require: $\{(X_i, Y_i)\}_{i=1}^n$, miscoverage level α

- 1: split n into n_0 training and n_1 calibration points
- 2: $\hat{q}_\tau(\cdot)$, $\tau \in \mathcal{G} \leftarrow$ TabPFN fitted through forward pass on n_0 training points
- 3: $t_i \leftarrow (7)$ for $i = n_0 + 1, \dots, n$ calibration points
- 4: $\hat{t} \leftarrow t_{(\lceil (1-\alpha)(n_1+1) \rceil)}$

Ensure: $\hat{C}_n(x)$ as in (6) with $t = \hat{t}$

3.4 CTabPFN-Q

CTabPFN-Q calibrates the quantile level itself, as DCP does in Section 2.2. It is convenient to index the family directly by the level τ ,

$$\hat{C}_{n_0, \tau}(x) = [\hat{q}_\tau(x), \hat{q}_{1-\tau}(x)], \quad \tau \in \mathcal{G} \cap (0, 0.5), \quad (8)$$

where the family grows as τ decreases. The score of a calibration point is the deepest level at which it is still covered,

$$\tau_i = \max \{ \tau \in \mathcal{G} \cap (0, 0.5) : \hat{q}_\tau(X_i) \leq Y_i \leq \hat{q}_{1-\tau}(X_i) \}, \quad (9)$$

with the convention $\tau_i = 0$ if no grid level covers Y_i , and the calibrated level is $\hat{\tau} = \tau_{(\lfloor \alpha(n_1+1) \rfloor)}$, the $\lfloor \alpha(n_1+1) \rfloor$ -th smallest score, or the smallest grid level if $\lfloor \alpha(n_1+1) \rfloor = 0$. Both endpoints of the returned interval are genuine quantiles of the posterior predictive, so the native shape is preserved exactly and only the nominal level is corrected. Unlike A and M, which need $\alpha/2 \in \mathcal{G}$, Q operates at any miscoverage level.

Algorithm 3 CTabPFN-Q

Require: $\{(X_i, Y_i)\}_{i=1}^n$, miscoverage level α

- 1: split n into n_0 training and n_1 calibration points
- 2: $\hat{q}_\tau(\cdot)$, $\tau \in \mathcal{G} \leftarrow$ TabPFN fitted through forward pass on n_0 training points
- 3: $\tau_i \leftarrow (9)$ for $i = n_0 + 1, \dots, n$ calibration points
- 4: $\hat{\tau} \leftarrow \tau_{(\lfloor \alpha(n_1+1) \rfloor)}$

Ensure: $\hat{C}_n(x) = [\hat{q}_{\hat{\tau}}(x), \hat{q}_{1-\hat{\tau}}(x)]$

3.5 Theoretical guarantees

Theorem 1. Assume the points $\{(X_i, Y_i)\}_{i=1}^{n+1}$ are exchangeable. Then the CTabPFN-A, CTabPFN-M, and CTabPFN-Q intervals of Algorithms 1 to 3 satisfy the marginal coverage guarantee (1).

Proof. The families (4), (6), and (8) are monotone in their index by the monotonicity of $\hat{q}_\tau$ in τ , and TabPFN is fit on the training points alone. Each method is therefore an instance of the nested construction of Section 2.1, and the guarantee follows from Vovk et al. (2005) and Gupta et al. (2022). $\square$

The following results quantify how far above $1 - \alpha$ the coverage can sit.

Proposition 1. Under the assumptions of Theorem 1, if the conformity scores are almost surely distinct, the CTabPFN-A and CTabPFN-M intervals also satisfy $\mathbb{P}(Y_{n+1} \in \hat{C}_n(X_{n+1})) \leq 1 - \alpha + \frac{1}{n_1+1}$.

Proof. The scores (5) and (7) are continuous in Y_i , so the standard upper bound for split conformal applies (Romano et al., 2019). $\square$

For CTabPFN-Q, the discreteness of $\mathcal{G}$ adds a small amount of conservatism. Rounding to the grid lowers each score (9) by at most the step $\delta = 0.001$, so $\hat{\tau}$ is at most δ smaller than on a continuous level. The nominal coverage $1 - 2\hat{\tau}$ of the returned interval is therefore at most $2\delta = 0.002$ higher than necessary. This bound is why we fix the step at 0.001, since an excess of at most 0.002 in coverage can be disregarded in most applications.

4 PERFORMANCE ON REAL DATA USING UQ-BENCH

4.1 UQ-Bench

All experiments run on **uq-bench**, our benchmark harness for uncertainty quantification.³ Its design follows the following three values:

1. Reproducibility Every run is deterministic and byte-identical given the seed, and each configuration is repeated over multiple seeds for stable results.

2. Encapsulation Every method lives in its own isolated environment with pinned dependencies, so methods cannot interfere with each other.

3. Fairness Every method receives exactly the same train–test split for each dataset and seed. If a calibration set is needed, the method splits it from the train data.

4.2 Experimental setup

Datasets We benchmark on 12 of the 20 datasets in the **uq-bench** hub, listed in Table 1. Each dataset is split 80/20 into train and test per seed, each configuration runs over 10 seeds at $\alpha = 0.1$, a nominal coverage of 90%.

³For more information about the benchmark and the harness, including our exact reproduction of published PCS-UQ and CLEAR results, consult Appendix A.

Table 1: Datasets used in the benchmark.

Dataset	n	p
energy_efficiency	768	10
concrete	1030	8
insurance	1338	8
airfoil	1503	5
debutanizer	2394	7
parkinsons	5875	18
computer	8192	21
kin8nm	8192	8
powerplant	9568	4
sulfur	10081	5
naval_propulsion	11934	24
miami_housing	13932	28

Methods We compare CTabPFN-A, CTabPFN-M, and CTabPFN-Q against the following four conformal baselines and native TabPFN:

- Baseline, split conformal regression on ordinary least squares with absolute-residual scores.
- UACQR-P, the authors’ quantile-random-forest recipe with $B = 100$.
- PCS-UQ, the authors’ implementation with $B = 1000$ bootstraps and top-1 model selection.
- CLEAR, the authors’ implementation with its quantile-based ensemble.
- Native TabPFN, the uncalibrated interval $[\hat{q}_{0.05}(x), \hat{q}_{0.95}(x)]$, a reference without a guarantee.

Metrics For test points $\{(X_i, Y_i)\}_{i=1}^m$ with interval bounds $\ell_i \leq u_i$, we report the prediction interval coverage probability

$$\text{PICP} = \frac{1}{m} \sum_{i=1}^m \mathbf{1}_{\ell_i \leq Y_i \leq u_i},$$

the normalized interval width

$$\text{NIW} = \frac{\frac{1}{m} \sum_{i=1}^m (u_i - \ell_i)}{\max_i Y_i - \min_i Y_i},$$

and the average interval score loss (Gneiting and Raftery, 2007; Rossellini et al., 2024)

$$\text{AISL} = \frac{1}{m} \sum_{i=1}^m \left[(u_i - \ell_i) + \frac{2}{\alpha} d_i \right],$$

where $d_i = (\ell_i - Y_i) \mathbf{1}_{Y_i < \ell_i} + (Y_i - u_i) \mathbf{1}_{Y_i > u_i}$ is the distance by which the interval misses Y_i . PICP should

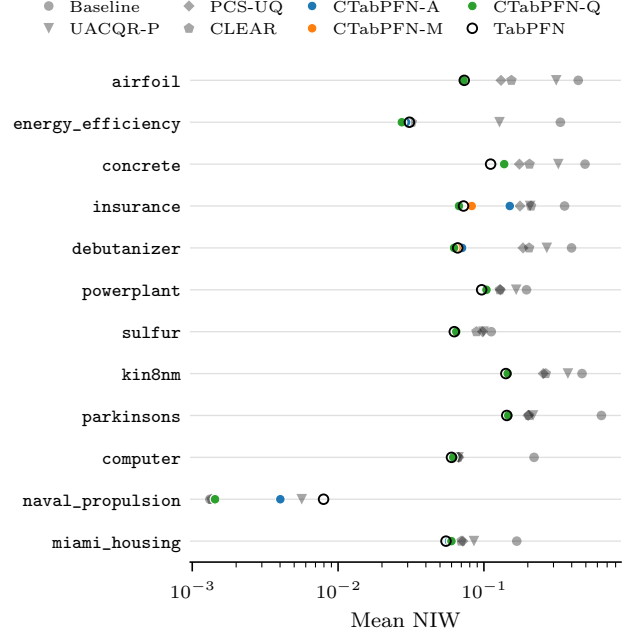


Figure 1: Mean NIW per dataset and method over 10 seeds, on a logarithmic axis. Datasets are ordered by $n \cdot p$ from top to bottom. Native TabPFN is drawn as an open circle.

reach the target $1 - \alpha$. Among methods that do, a smaller NIW is better. AISL is a proper scoring rule balancing width against miscoverage, and we normalize it by the test response range when averaging across datasets.

4.3 Results

Table 2 shows that every conformal method holds the target coverage on every dataset up to seed noise, while native TabPFN swings between 0.875 and 1.000. The lowest conformal row, **energy_efficiency**, is shared by all conformal methods on the same splits. It is consistent with finite-sample variation on the smallest dataset, since the guarantee holds in expectation over the split.

Table 3 and Figure 1 show that on 11 of 12 datasets, the narrowest interval with a guarantee comes from a CTabPFN variant, with a multi-way tie at the table’s precision on **naval_propulsion**. The margins over the retrained baselines are large. On **airfoil**, for instance, the CTabPFN intervals are roughly half the PCS-UQ width and a third of the UACQR-P width. Against native TabPFN the calibrated intervals give up little and sometimes gain. On **energy_efficiency**, **airfoil**, **debutanizer**, and **naval_propulsion** CTabPFN-M or CTabPFN-Q is

Table 2: Per-dataset PICP over 10 seeds at nominal coverage 0.9. Subscripts give the standard deviation across seeds in units of 10^{-2} . A, M, and Q are short for CTabPFN-A, CTabPFN-M, and CTabPFN-Q.

Dataset	Baseline	UACQR-P	PCS-UQ	CLEAR	A	M	Q	TabPFN
energy_efficiency	0.913 _{2.7}	0.899 _{1.7}	0.888 _{2.2}	0.882 _{2.2}	0.892 _{3.2}	0.884 _{3.7}	0.882 _{4.2}	0.939 _{2.4}
concrete	0.907 _{4.0}	0.908 _{1.8}	0.893 _{2.7}	0.905 _{2.7}	0.907 _{3.8}	0.901 _{3.5}	0.900 _{3.9}	0.875 _{2.2}
insurance	0.913 _{1.7}	0.909 _{1.6}	0.895 _{1.3}	0.894 _{1.9}	0.893 _{2.9}	0.897 _{2.5}	0.897 _{2.3}	0.910 _{1.1}
airfoil	0.887 _{3.0}	0.907 _{2.2}	0.905 _{1.9}	0.899 _{2.0}	0.894 _{3.3}	0.890 _{3.8}	0.893 _{3.7}	0.914 _{1.9}
debutanizer	0.904 _{2.0}	0.894 _{2.0}	0.891 _{1.5}	0.897 _{1.7}	0.903 _{1.8}	0.901 _{2.1}	0.908 _{1.6}	0.939 _{1.1}
parkinsons	0.902 _{1.1}	0.903 _{1.0}	0.904 _{0.7}	0.899 _{1.5}	0.902 _{2.3}	0.902 _{2.1}	0.905 _{2.1}	0.895 _{1.2}
computer	0.899 _{1.0}	0.905 _{0.7}	0.901 _{0.6}	0.901 _{1.0}	0.902 _{1.5}	0.901 _{1.6}	0.902 _{1.8}	0.898 _{0.8}
kin8nm	0.906 _{0.8}	0.909 _{1.2}	0.904 _{1.1}	0.905 _{1.4}	0.899 _{1.5}	0.900 _{1.6}	0.901 _{1.6}	0.891 _{1.0}
powerplant	0.900 _{1.0}	0.905 _{0.7}	0.908 _{0.8}	0.902 _{1.2}	0.897 _{1.2}	0.897 _{1.1}	0.898 _{1.2}	0.886 _{0.7}
sulfur	0.904 _{0.7}	0.903 _{1.0}	0.905 _{0.7}	0.900 _{1.5}	0.904 _{1.1}	0.904 _{1.1}	0.905 _{1.2}	0.907 _{0.5}
naval_propulsion	0.900 _{0.9}	0.907 _{1.1}	0.904 _{0.5}	0.900 _{0.7}	0.892 _{1.0}	0.897 _{0.8}	0.899 _{1.2}	1.000 _{0.0}
miami_housing	0.903 _{0.6}	0.903 _{0.8}	0.904 _{0.5}	0.901 _{1.0}	0.898 _{0.8}	0.898 _{0.8}	0.899 _{0.8}	0.876 _{0.9}

 Table 3: Per-dataset NIW over 10 seeds. Subscripts give the standard deviation across seeds in units of 10^{-2} . A, M, and Q are short for CTabPFN-A, CTabPFN-M, and CTabPFN-Q. Highlighted cells mark the narrowest interval among the methods with a coverage guarantee.

Dataset	Baseline	UACQR-P	PCS-UQ	CLEAR	A	M	Q	TabPFN
energy_efficiency	0.335 _{2.0}	0.128 _{1.0}	0.030 _{0.2}	0.032 _{0.4}	0.029 _{0.2}	0.027 _{0.3}	0.027 _{0.3}	0.031 _{0.2}
concrete	0.494 _{3.2}	0.324 _{1.4}	0.175 _{0.6}	0.205 _{2.2}	0.136 _{1.5}	0.138 _{2.1}	0.138 _{2.1}	0.111 _{0.8}
insurance	0.357 _{4.6}	0.207 _{3.5}	0.177 _{5.9}	0.210 _{6.5}	0.150 _{10.8}	0.082 _{5.9}	0.067 _{7.9}	0.072 _{1.4}
airfoil	0.443 _{2.6}	0.313 _{3.1}	0.131 _{0.6}	0.154 _{1.6}	0.074 _{0.7}	0.072 _{0.9}	0.072 _{1.0}	0.073 _{0.4}
debutanizer	0.399 _{1.1}	0.270 _{1.4}	0.185 _{0.5}	0.204 _{1.0}	0.071 _{0.5}	0.065 _{0.7}	0.062 _{0.8}	0.066 _{0.5}
parkinsons	0.639 _{2.4}	0.217 _{1.1}	0.201 _{0.2}	0.203 _{0.5}	0.145 _{0.3}	0.146 _{0.4}	0.147 _{0.4}	0.143 _{0.2}
computer	0.221 _{0.9}	0.067 _{0.2}	0.067 _{0.5}	0.067 _{0.4}	0.061 _{0.2}	0.061 _{0.3}	0.062 _{0.3}	0.060 _{0.1}
kin8nm	0.471 _{0.8}	0.377 _{0.9}	0.255 _{0.5}	0.266 _{0.7}	0.145 _{0.4}	0.145 _{0.4}	0.146 _{0.5}	0.141 _{0.3}
powerplant	0.196 _{0.9}	0.166 _{0.8}	0.130 _{0.5}	0.129 _{0.5}	0.103 _{0.5}	0.103 _{0.5}	0.104 _{0.6}	0.097 _{0.3}
sulfur	0.112 _{0.8}	0.099 _{0.8}	0.098 _{0.7}	0.089 _{0.7}	0.064 _{0.5}	0.064 _{0.5}	0.065 _{0.5}	0.063 _{0.5}
naval_propulsion	0.001 _{0.0}	0.006 _{0.1}	0.001 _{0.0}	0.001 _{0.0}	0.004 _{0.0}	0.001 _{0.0}	0.001 _{0.0}	0.008 _{0.0}
miami_housing	0.168 _{0.3}	0.085 _{0.2}	0.072 _{0.3}	0.070 _{0.4}	0.058 _{0.2}	0.060 _{0.2}	0.060 _{0.2}	0.055 _{0.2}

narrower than the native interval while restoring the target level. The AISL results in Appendix B show the same ordering under a proper scoring rule on 11 of 12 datasets. The exception is **insurance**, where the miss-distance penalty favors UACQR-P. Per dataset, the width of the best CTabPFN variant relative to the best retrained baseline ranges from parity on **naval_propulsion** to a factor of three on **debutanizer**. Across datasets, the median width of the best CTabPFN variant is 0.75 of the best retrained baseline.

Two rows separate the three variants. On **insurance**, a dataset with a heavy-tailed target, CTabPFN-A widens to 0.150 while CTabPFN-M and CTabPFN-Q read 0.082 and 0.067. As anticipated in Section 3.2, the additive margin is driven by the largest calibration errors, while the relative scores are not. On **naval_propulsion**, native TabPFN overcovers at 1.000, and the calibrated methods shrink their interval below the native one by a factor of eight, from an NIW of 0.008 to 0.001, which is the case $\hat{t} < 1$ of Section 3.3.

5 ABLATION STUDIES

The training fraction n_0/n trades interval sharpness against calibration stability. A larger n_0 gives TabPFN more context and hence a better posterior, while a smaller n_1 leaves fewer points to calibrate on. We sweep the fraction over $\{0.5, 0.6, 0.7, 0.75, 0.8, 0.9, 0.95\}$ on all 12 datasets and 10 seeds, running all three variants at each cell.

Table 4 summarizes the sweep. Mean coverage is invariant in the fraction, as the guarantee requires. What changes are the spread and the width. The per-cell standard deviation of coverage grows from 0.016 at fraction 0.5 to 0.031 at 0.95, and the width falls monotonically as TabPFN receives more context. Figure 2 shows four representative datasets.

On this evidence we set the default fraction to 0.9, where the width gain is 3.3% over 0.75 at a still moderate spread. The choice also follows from the nature of the model. TabPFN infers in context, so its posterior improves with every training point it sees, while

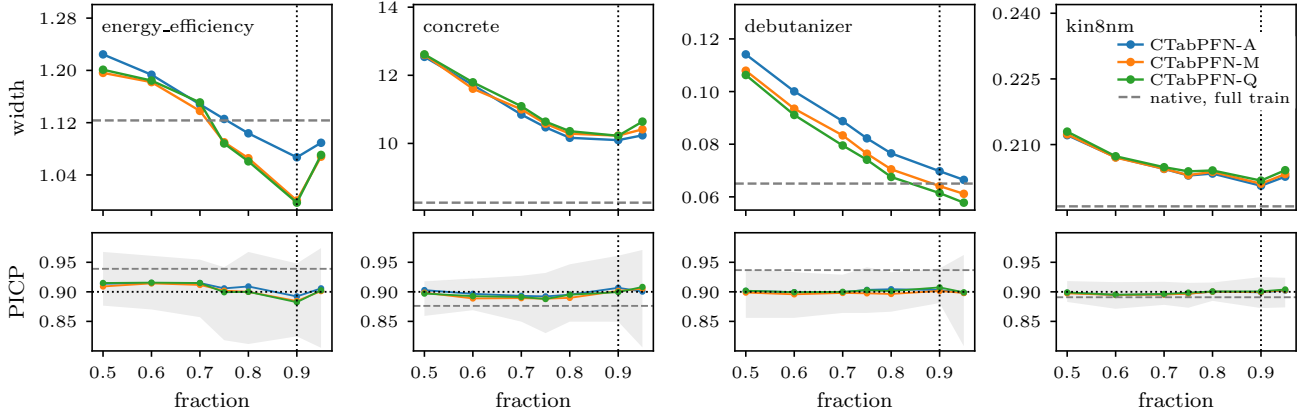


Figure 2: Interval width and test coverage against the training fraction on four datasets, 10 seeds. The gray band pools the per-seed coverage range over the three variants, the dashed line is native TabPFN at full train context, the dotted vertical line marks the default fraction 0.9.

Table 4: Ablation of the training fraction, pooled over datasets, variants, and seeds. Width ratio is the median interval width relative to the fraction 0.75.

Fraction	PICP	sd(PICP)	Width ratio
0.50	0.900	0.016	1.078
0.60	0.899	0.016	1.037
0.70	0.899	0.017	1.011
0.75	0.898	0.019	1.000
0.80	0.899	0.019	0.986
0.90	0.898	0.023	0.967
0.95	0.900	0.031	0.965

calibrating a single parameter needs only on the order of $1/\alpha$ points. We disclose that this choice was made after seeing the sweep on the four datasets of Figure 2, and it was confirmed on the remaining eight. The residual risk is confined to small datasets. There a fraction of 0.9 leaves few calibration points, for example only 61 on **energy_efficiency**, and the per-seed coverage fluctuates accordingly.

6 DISCUSSION

We have shown that the predictive distribution TabPFN natively provides is a strong base for conformal regression. Calibrating it restores the finite-sample guarantee at almost no width cost. The resulting intervals are substantially narrower than those of conformal methods that train their own uncertainty models. The three variants perform similarly across our benchmark, and none dominates. CTabPFN-Q offers two practical conveniences, since it preserves the native shape exactly and operates at any miscoverage

level, while CTabPFN-A was the narrowest variant on several datasets.

6.1 Future work

Future work is laid out as follows.

- A held-out validation on the six **uq-bench** datasets untouched by any of our choices.
- A synthetic dataset study of conditional coverage, comparing the methods where the true conditional quantiles are known.
- Reporting further metrics that **uq-bench** already computes, such as the pinball loss and calibrated interval widths.
- An ablation over the coverage level beyond $\alpha = 0.1$.
- A more adaptive rule for the training fraction n_0/n , aware of the sample size and dimension of the dataset at hand.
- Removing the grid in CTabPFN-Q by calibrating in the probability integral transform of TabPFN’s full native CDF.
- Calibrating the two tails separately for a stronger, asymmetric guarantee (Romano et al., 2019).
- Extending the construction to classification through the class probabilities of TabPFN, and growing **uq-bench** in methods, datasets, and metrics toward a community standard.

We note that such studies are compute intensive. Due to the repeated-run nature of the benchmark, a full

sweep over methods, datasets, and seeds takes a day to finish even on a compute cluster.

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A THE UQ-BENCH HARNESS

Adapters Every method is an adapter around its authors’ published implementation. Each adapter lives in its own project with a pinned, isolated environment. It communicates with the harness through a minimal file contract, since it reads a split of train and test arrays and writes interval bounds. The harness evaluates all bounds with the same metric code.

Reproducibility Results are stored in a keyed table, which makes every run resumable. A test suite asserts that repeated runs are byte-identical.

Inference cache The TabPFN family shares a content-addressed inference cache. One forward pass per dataset and seed serves the native reference and all three CTabPFN variants at every coverage level.

Dataset hub The datasets live in a public hub alongside the harness. Each dataset is a single table with a fixed target column, downloaded and cached on first use, and loaded identically for every method. Twenty regression datasets are available, of which this paper uses twelve, with six held out untouched for a future confirmatory evaluation.

Replication We validate the adapters by reproducing published results on the authors’ own splits and data. For CLEAR, our runs match the archived per-seed results exactly, to all six printed decimals in 30 of 30 cells. For PCS-UQ, coverage matches exactly in 30 of 30 cells and interval widths agree to the fourth decimal once the environment pins the authors’ solver and scikit-learn version, with the residual difference attributable to a newer xgboost.

Extensibility and performance Adding a method requires one folder, which holds its own environment specification and a small entry script implementing the file contract. The harness discovers the folder, schedules its cells alongside the others, and scores its output with the shared metrics. Per-method thread and worker counts are tuned so that a full sweep saturates the machine, with independent cells running in parallel, scheduled largest-first. Completed cells are never recomputed, so a repeated sweep replays from the results table and the inference cache within minutes.

B ADDITIONAL RESULTS

The following table complements Section 4.3 with the per-dataset AISL.

Table 5: Per-dataset AISL normalized by the test response range, mean over 10 seeds. A, M, and Q are short for CTabPFN-A, CTabPFN-M, and CTabPFN-Q. Highlighted cells mark the best value among the methods with a coverage guarantee.

Dataset	Baseline	UACQR-P	PCS-UQ	CLEAR	A	M	Q	TabPFN
energy_efficiency	0.383	0.148	0.045	0.051	0.039	0.039	0.039	0.039
concrete	0.604	0.407	0.254	0.287	0.198	0.200	0.199	0.189
insurance	0.495	0.402	0.425	0.417	0.459	0.458	0.432	0.453
airfoil	0.626	0.386	0.180	0.217	0.107	0.108	0.108	0.100
debutanizer	0.664	0.362	0.256	0.283	0.094	0.094	0.092	0.084
parkinsons	0.842	0.254	0.259	0.270	0.173	0.173	0.173	0.172
computer	0.443	0.107	0.092	0.093	0.082	0.082	0.082	0.081
kin8nm	0.595	0.454	0.328	0.336	0.181	0.181	0.181	0.180
powerplant	0.253	0.218	0.174	0.179	0.153	0.153	0.153	0.149
sulfur	0.244	0.142	0.128	0.131	0.085	0.085	0.085	0.082
naval_propulsion	0.002	0.006	0.002	0.002	0.004	0.002	0.002	0.008
miami_housing	0.287	0.120	0.094	0.099	0.083	0.083	0.083	0.082

C ADDENDUM: TABPFN 3.5

TabPFN 3.5 was released on 2026-09-15, after the benchmark in Section 4 was run. This addendum repeats the TabPFN family of methods on the new generation. The v3.5 methods pin `ctabpfm` v0.2.0, which pins `tabpfm` 9.0.0 with the v3.5 weights, while the v3 methods of the main text stay pinned to v0.1.0 on `tabpfm` 8.5.0. Splits, seeds, coverage level, and metrics are unchanged, and the four retrained baselines are not rerun. Tables 6 to 8 report per-dataset PICP, NIW, and normalized AISL for both generations side by side.

Native TabPFN 3.5 covers more than native 3 on every dataset. Native v3 undercovered the target on 4 of the 12 datasets, `concrete`, `miami_housing`, `powerplant`, and `kin8nm`, while native 3.5 undercovers on none of them and sits between 0.905 and 0.946, at roughly unchanged width. All six calibrated columns hold the target coverage within seed noise on both generations, and calibrated on 3.5 the narrowest variant is narrower than on v3 on 7 of 12 datasets.

The picture under AISL, the proper scoring rule, is less favorable for calibration. Native 3.5 attains the lowest AISL on 10 of 12 datasets, the exceptions being `energy_efficiency` and `naval_propulsion`. The margins are within 0.002 on most datasets and reach 0.008 to 0.011 on `concrete`, `insurance`, and `debutanizer`. Two effects combine here. AISL penalizes a miss at twenty times its distance while charging width linearly, so the mild overcoverage of native 3.5 is cheap and the calibrated intervals, pulled to exactly 0.9, pay for their additional misses. In addition, the calibrated variants see 90% of the training points while native sees all of them. Since overcoverage is rarely harmful in practice, the practical benefit of conformalizing a base model that is already conservative is smaller than it was for v3, and reduces to the guarantee itself, which native 3.5 does not carry on unseen datasets.

Table 6: Per-dataset PICP over 10 seeds at nominal coverage 0.9 for the TabPFN family on v3 and v3.5. Subscripts give the standard deviation across seeds in units of 10^{-2} . A, M, and Q are short for CTabPFN-A, CTabPFN-M, and CTabPFN-Q, the two rightmost columns are native TabPFN.

Dataset	CTabPFN v3			CTabPFN v3.5			v3	v3.5
	A	M	Q	A	M	Q	TabPFN	TabPFN
<code>energy_efficiency</code>	0.892 _{3.2}	0.884 _{3.7}	0.882 _{4.2}	0.895 _{5.2}	0.894 _{5.0}	0.894 _{5.3}	0.939 _{2.4}	0.940 _{2.4}
<code>concrete</code>	0.907 _{3.8}	0.901 _{3.5}	0.900 _{3.9}	0.887 _{3.8}	0.883 _{4.5}	0.883 _{4.7}	0.875 _{2.2}	0.890 _{2.4}
<code>insurance</code>	0.893 _{2.9}	0.897 _{2.5}	0.897 _{2.3}	0.904 _{3.0}	0.903 _{2.8}	0.901 _{2.7}	0.910 _{1.1}	0.926 _{1.4}
<code>airfoil</code>	0.894 _{3.3}	0.890 _{3.8}	0.893 _{3.7}	0.896 _{3.7}	0.901 _{3.2}	0.903 _{2.9}	0.914 _{1.9}	0.930 _{1.3}
<code>debutanizer</code>	0.903 _{1.8}	0.901 _{2.1}	0.908 _{1.6}	0.903 _{2.5}	0.904 _{2.1}	0.907 _{2.4}	0.939 _{1.1}	0.946 _{0.9}
<code>parkinsons</code>	0.902 _{2.3}	0.902 _{2.1}	0.905 _{2.1}	0.899 _{1.9}	0.899 _{2.0}	0.901 _{2.1}	0.895 _{1.2}	0.902 _{1.3}
<code>computer</code>	0.902 _{1.5}	0.901 _{1.6}	0.902 _{1.8}	0.905 _{1.8}	0.905 _{2.0}	0.906 _{1.9}	0.898 _{0.8}	0.917 _{0.7}
<code>kin8nm</code>	0.899 _{1.5}	0.900 _{1.6}	0.901 _{1.6}	0.902 _{1.6}	0.902 _{1.6}	0.902 _{1.6}	0.891 _{1.0}	0.911 _{0.8}
<code>powerplant</code>	0.897 _{1.2}	0.897 _{1.1}	0.898 _{1.2}	0.900 _{1.0}	0.900 _{1.0}	0.900 _{1.1}	0.886 _{0.7}	0.908 _{0.8}
<code>sulfur</code>	0.904 _{1.1}	0.904 _{1.1}	0.905 _{1.2}	0.899 _{1.4}	0.902 _{1.3}	0.901 _{1.4}	0.907 _{0.5}	0.937 _{0.5}
<code>naval_propulsion</code>	0.892 _{1.0}	0.897 _{0.8}	0.899 _{1.2}	0.890 _{1.3}	0.886 _{1.1}	0.894 _{1.3}	1.000 _{0.0}	1.000 _{0.0}
<code>miami_housing</code>	0.898 _{0.8}	0.898 _{0.8}	0.899 _{0.8}	0.899 _{0.8}	0.899 _{0.8}	0.900 _{0.7}	0.876 _{0.9}	0.905 _{0.6}

Table 7: Per-dataset NIW over 10 seeds for the TabPFN family on v3 and v3.5. Subscripts give the standard deviation across seeds in units of 10^{-2} . Highlighted cells mark the narrowest interval among the six methods with a coverage guarantee.

Dataset	CTabPFN v3			CTabPFN v3.5			v3	v3.5
	A	M	Q	A	M	Q	TabPFN	TabPFN
energy_efficiency	0.029 _{0.2}	0.027 _{0.3}	0.027 _{0.3}	0.030 _{0.2}	0.029 _{0.3}	0.029 _{0.3}	0.031 _{0.2}	0.032 _{0.1}
concrete	0.136 _{1.5}	0.138 _{2.1}	0.138 _{2.1}	0.123 _{1.4}	0.119 _{1.7}	0.120 _{2.0}	0.111 _{0.8}	0.110 _{0.8}
insurance	0.150 _{10.8}	0.082 _{5.9}	0.067 _{7.9}	0.149 _{2.2}	0.117 _{2.7}	0.071 _{6.4}	0.072 _{1.4}	0.145 _{2.5}
airfoil	0.074 _{0.7}	0.072 _{0.9}	0.072 _{1.0}	0.077 _{0.7}	0.076 _{0.8}	0.076 _{0.8}	0.073 _{0.4}	0.078 _{0.3}
debutanizer	0.071 _{0.5}	0.065 _{0.7}	0.062 _{0.8}	0.053 _{0.3}	0.046 _{0.6}	0.044 _{0.7}	0.066 _{0.5}	0.048 _{0.3}
parkinsons	0.145 _{0.3}	0.146 _{0.4}	0.147 _{0.4}	0.146 _{0.3}	0.146 _{0.3}	0.146 _{0.4}	0.143 _{0.2}	0.145 _{0.2}
computer	0.061 _{0.2}	0.061 _{0.3}	0.062 _{0.3}	0.059 _{0.1}	0.058 _{0.1}	0.056 _{0.3}	0.060 _{0.1}	0.058 _{0.1}
kin8nm	0.145 _{0.4}	0.145 _{0.4}	0.146 _{0.5}	0.145 _{0.4}	0.145 _{0.4}	0.145 _{0.4}	0.141 _{0.3}	0.147 _{0.3}
powerplant	0.103 _{0.5}	0.103 _{0.5}	0.104 _{0.6}	0.091 _{0.4}	0.090 _{0.4}	0.090 _{0.4}	0.097 _{0.3}	0.092 _{0.4}
sulfur	0.064 _{0.5}	0.064 _{0.5}	0.065 _{0.5}	0.064 _{0.5}	0.061 _{0.5}	0.057 _{0.4}	0.063 _{0.5}	0.064 _{0.5}
naval_propulsion	0.004 _{0.0}	0.001 _{0.0}	0.001 _{0.0}	0.003 _{0.0}	0.001 _{0.0}	0.001 _{0.0}	0.008 _{0.0}	0.004 _{0.0}
miami_housing	0.058 _{0.2}	0.060 _{0.2}	0.060 _{0.2}	0.057 _{0.2}	0.057 _{0.2}	0.057 _{0.2}	0.055 _{0.2}	0.057 _{0.2}

Table 8: Per-dataset AISL normalized by the test response range, mean over 10 seeds, for the TabPFN family on v3 and v3.5. Highlighted cells mark the best value among the six methods with a coverage guarantee.

Dataset	CTabPFN v3			CTabPFN v3.5			v3	v3.5
	A	M	Q	A	M	Q	TabPFN	TabPFN
energy_efficiency	0.039	0.039	0.039	0.042	0.042	0.042	0.039	0.041
concrete	0.198	0.200	0.199	0.183	0.185	0.186	0.189	0.175
insurance	0.459	0.458	0.432	0.389	0.396	0.422	0.453	0.378
airfoil	0.107	0.108	0.108	0.108	0.107	0.107	0.100	0.100
debutanizer	0.094	0.094	0.092	0.069	0.071	0.067	0.084	0.059
parkinsons	0.173	0.173	0.173	0.168	0.168	0.169	0.172	0.166
computer	0.082	0.082	0.082	0.077	0.077	0.077	0.081	0.076
kin8nm	0.181	0.181	0.181	0.178	0.178	0.178	0.180	0.177
powerplant	0.153	0.153	0.153	0.138	0.138	0.138	0.149	0.137
sulfur	0.085	0.085	0.085	0.080	0.080	0.079	0.082	0.077
naval_propulsion	0.004	0.002	0.002	0.003	0.002	0.002	0.008	0.004
miami_housing	0.083	0.083	0.083	0.079	0.079	0.079	0.082	0.078